

CodeMentor AI

Explain. Debug. Learn. Grow.

Generative AI-Powered Coding Learning Platform

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Live Demo: <https://wist10.lovable.app>

INTRODUCTION

CodeMentor AI is an AI-powered coding learning platform designed to help students understand programming in a smarter and easier way.

It helps users to:

- Explain code
- Debug errors
- Analyze complexity
- Simplify code
- Practice coding
- Prepare for interviews
- Learn through voice support



PROBLEM STATEMENT

Many students face difficulties in:

1. Understanding programming logic
2. Fixing code errors
3. Learning multiple programming languages
4. Preparing for coding interviews
5. Practicing coding in a guided way

Existing tools are often either too complex or too limited for beginner-friendly learning

Objectives

The main objective of CodeMentor AI is to build a smart educational platform that helps students:

Objective 1

Learn
programming
interactively

Objective 2

Understand
code clearly

Objective 3

Improve
problem-
solving skills

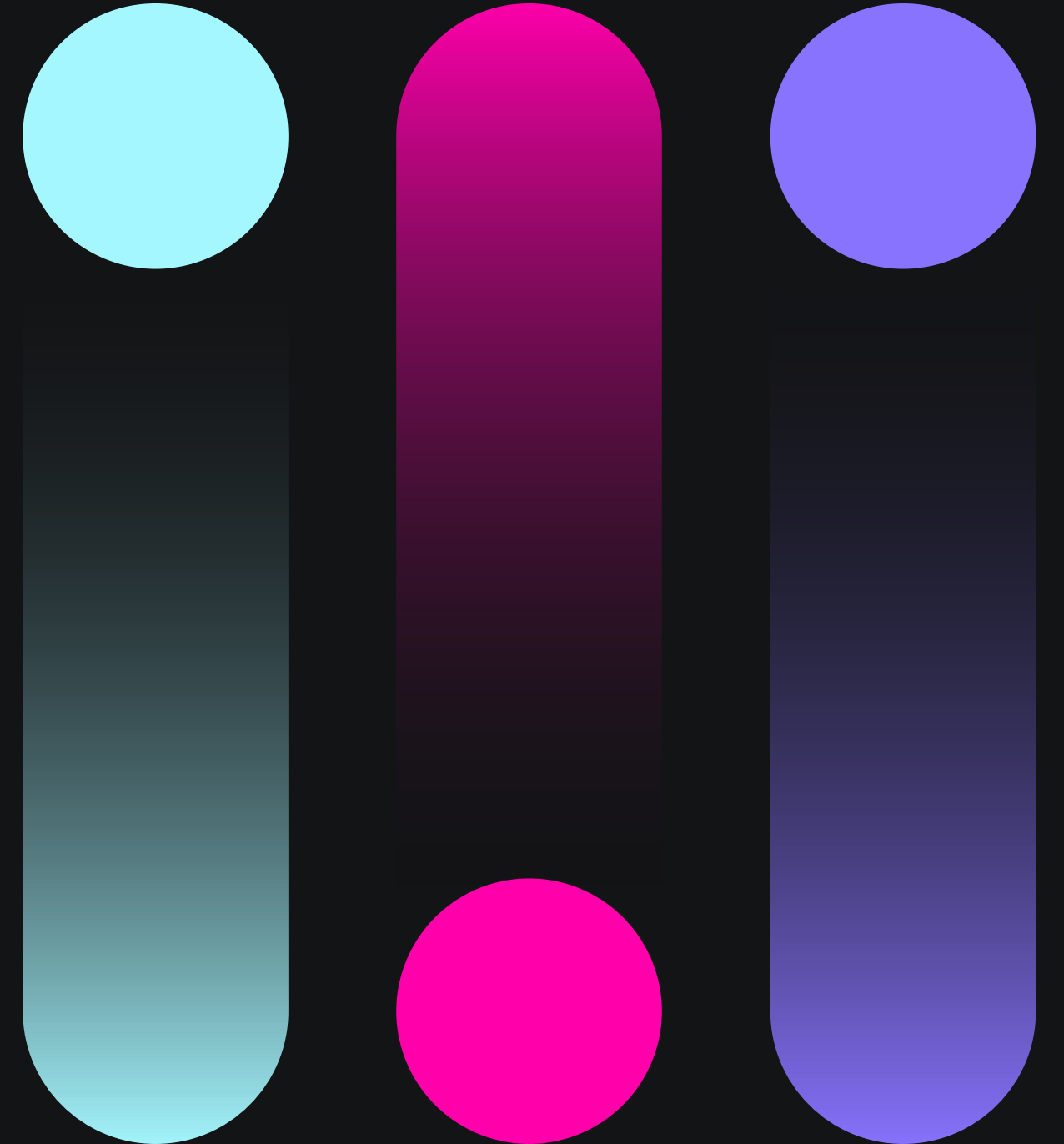
Objective 4

Get AI-based
coding
guidance

PROPOSED SOLUTION

We developed CodeMentor AI,
a web-based AI platform that allows users to:

- *Select a programming language*
- *Paste or enter code*
- *Choose a feature*
- *Receive AI-generated learning*



TECHNOLOGIES USED

Technologies Used Content:

Frontend:
Lovable
HTML
CSS
JavaScript

Backend / AI Integration:

API Integration
Generative AI Tools
Prompt Engineering

Other:

UI/UX Design

KEY FEATURES

- **Code Explanation**
- **AI Debugger**
- **Complexity Analyzer**
- **Code Simplifier**
- **Practice Generator**
- **Daily Coding Challenge**
- **Interview / Viva Questions**
- **Voice Companion**

SUPPORTED LANGUAGES

Our platform currently supports:

JavaScript

Python

Java

C

C++

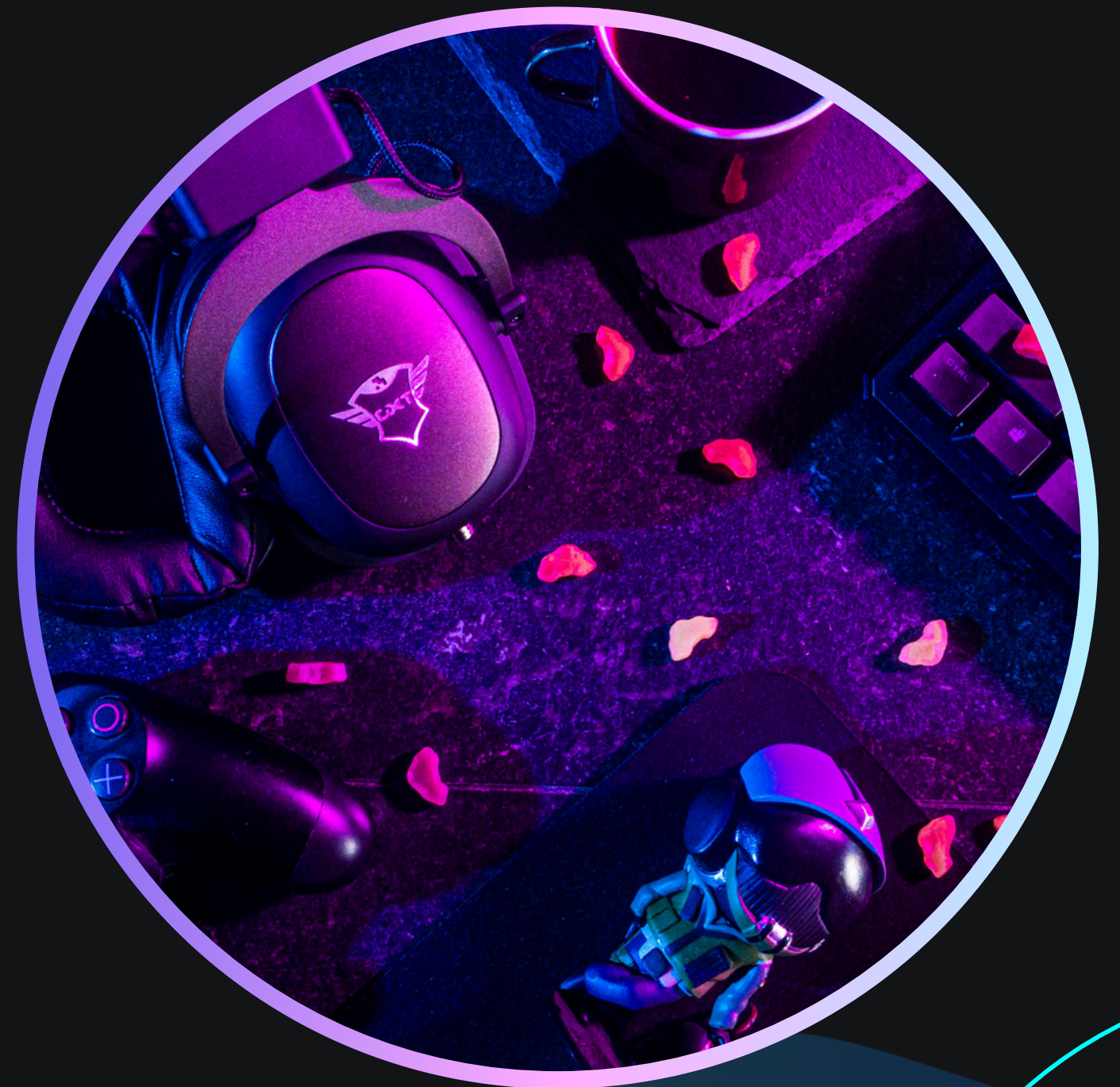
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CSS

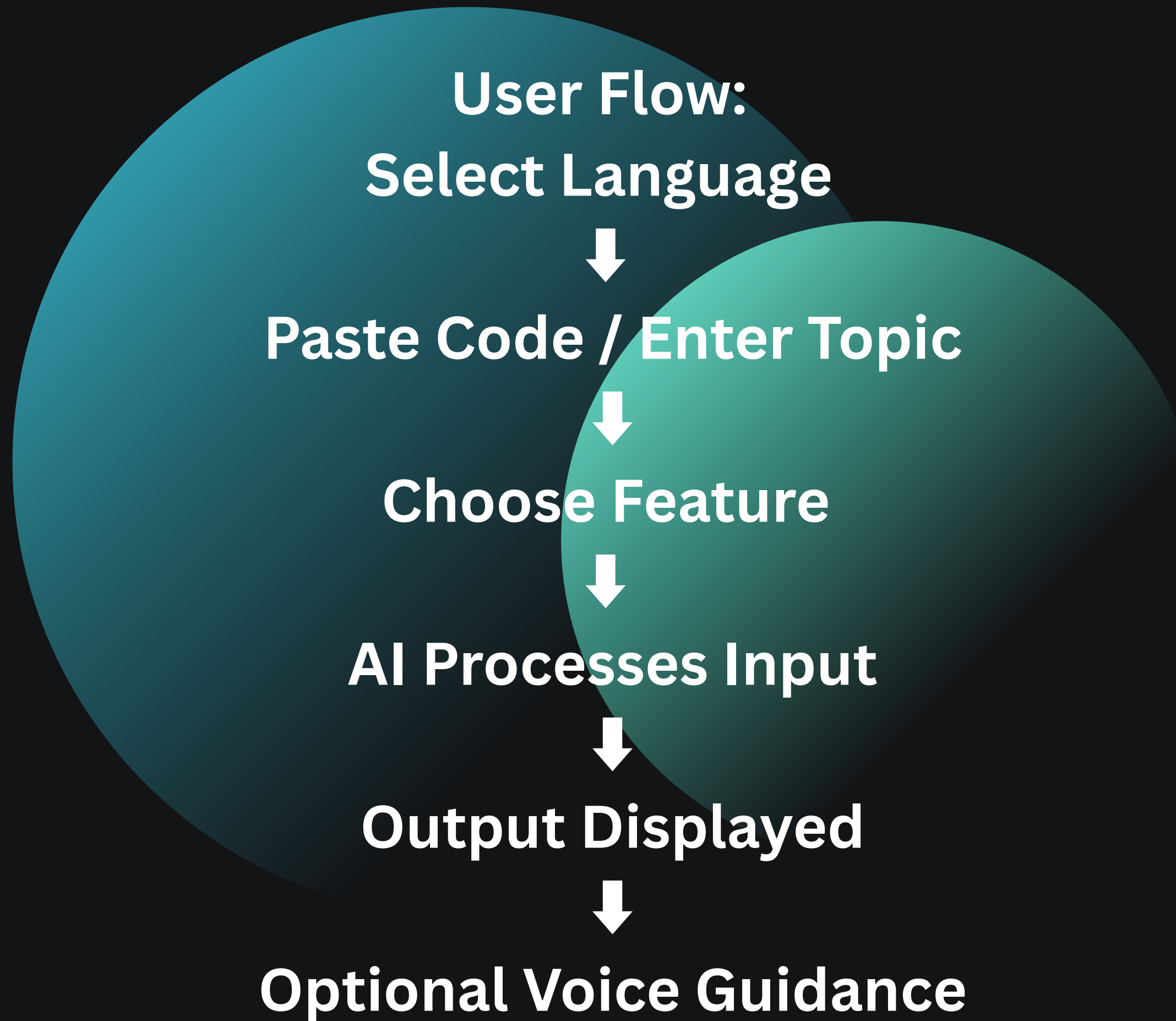
SQL

PHP

R



SYSTEM WORKFLOW



Badge System & Achievements

Adds fun and motivation to learning
Helps track student progress

Badge Levels:

- Bronze Explorer – First use
- Silver Learner – Uses multiple features
- Gold Coder – Regular practice
- Platinum Master – High engagement

Purpose:

Encourages continuous learning
Boosts engagement
Builds confidence

LEARNING OUTCOMES

Through this project, we gained practical experience in:

Generative AI integration

Prompt-based feature design

Frontend and backend understanding

API usage

Real-world problem solving

Team collaboration

CONCLUSION

CodeMentor AI is not just a code explainer — it is a complete AI-powered coding learning assistant.

Thank You

